Step 5 : Calculation of test statistic

The value of chi-square under H_0 is calculated as

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{19 \times 0.024}{0.018} = 25.3$$

Step 6 : **Critical value**

Since H_1 is a one-sided alternative, the critical values at α =0.05 is $\chi_e^2 = \chi^2_{19,\,0.95} = 10.117$.

Step 7 : **Decision**

Since it is a one-tailed test, the elements of the critical region are determined by the rejection rule $\chi_0^2 < \chi_e^2$

For the given sample information, the rejection rule does not hold, since

$\chi_0^2 = 25.3 > \chi_e^2 = \chi^2_{19,\,0.95} = 10.117$.

Hence, H_0 is not rejected in favour of H_1. Thus, the population variance can be regarded as 0.018.

2.2.5 Test of Hypotheses for independence of Attributes

Another important application of χ^2 test is the testing of independence of attributes.

Attributes: Attributes are qualitative characteristic such as levels of literacy, employment status, *etc*., which are quantified in terms of levels/scores.

Contigency table: Independence of two attributes is an important statistical application in which the data pertaining to the attributes are cross classified in the form of a two – dimensional table. The levels of one attribute are arranged in rows and of the other in columns. Such an arrangement in the form of a table is called as a contingency table.

Computational steps for testing the independence of attributes:

Step 1 : **Framing the hypotheses**

Null hypothesis H_0: The two attributes are independent

Alternative hypothesis H_1: The two attributes are not independent.

Step 2 : **Data**

The data set is given in the form of a contigency as under. Compute expected frequencies E_{ij} corresponding to each cell of the contingency table, using the formula

$$E_{ij} = \frac{R_i \times C_j}{N};\ i = 1, 2, \ldots m;\ j = 1, 2, \ldots n$$

where,

N = Total sample size

R_i = Row sum corresponding to i^{th} row

C_j = Column sum corresponding to j^{th} column

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